

Task 4

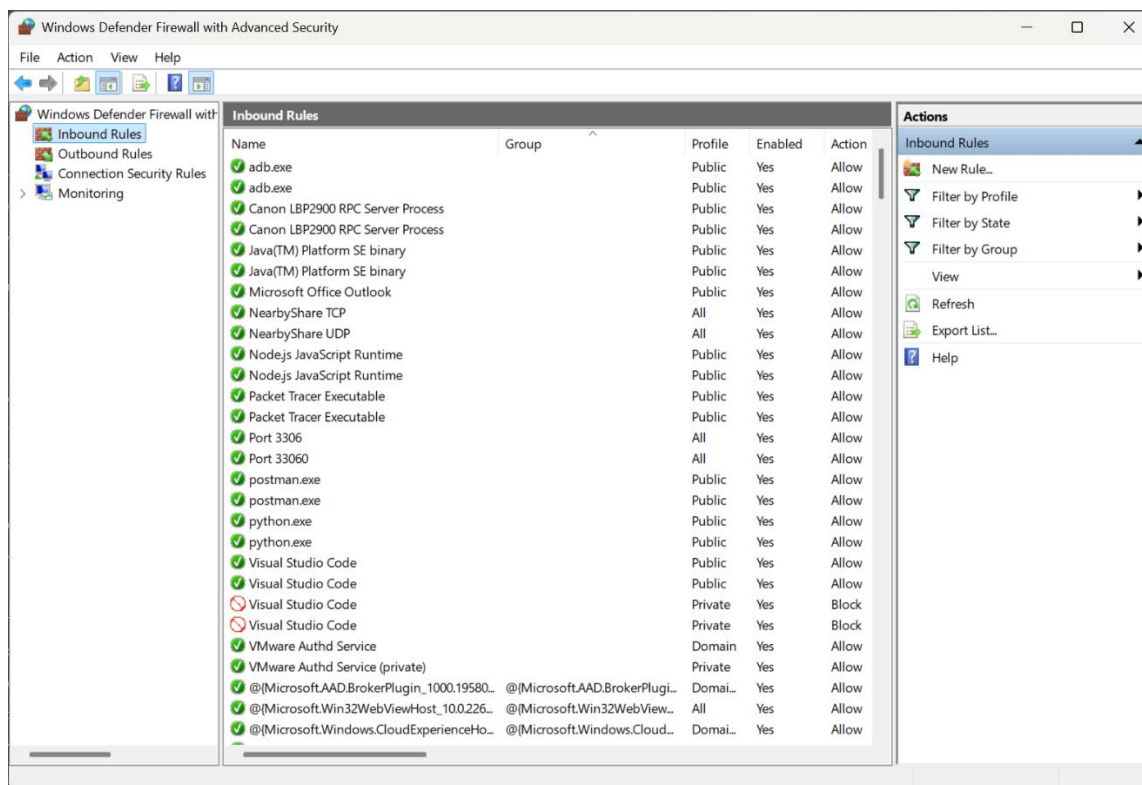
Setup and Use a Firewall on Windows/Linux

Objective: Configure and test basic firewall rules to allow or block traffic.

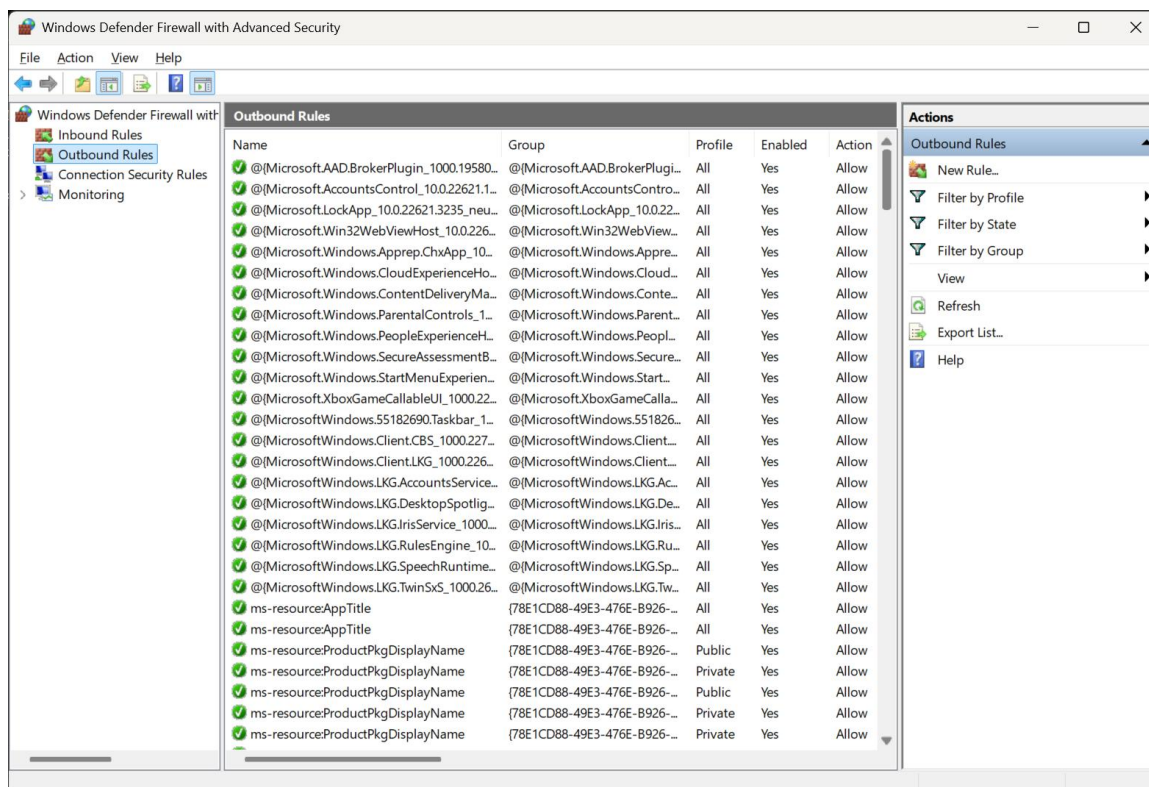
1.Open firewall configuration tool.

2.List current firewall rules.

These are the Inbound Rules.



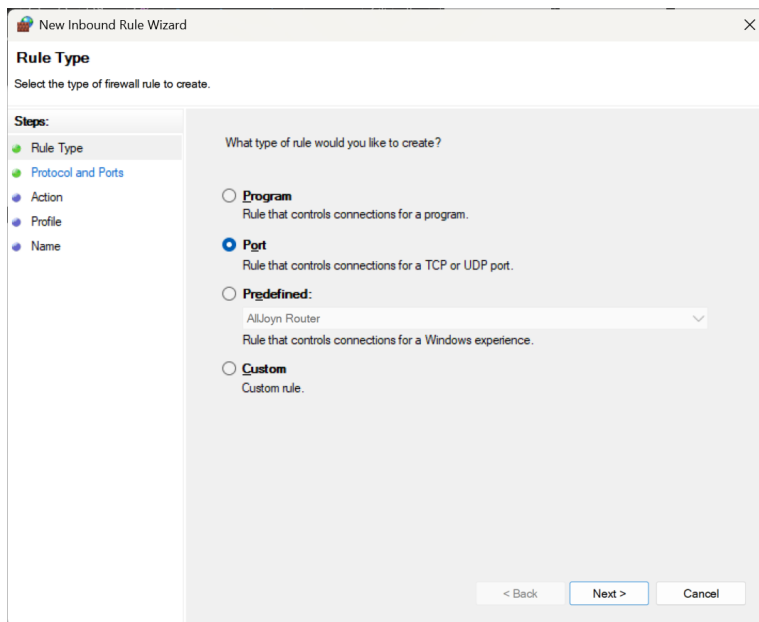
Here's a list of Outbound Rules.



3. Add a rule to block inbound traffic on a specific port.

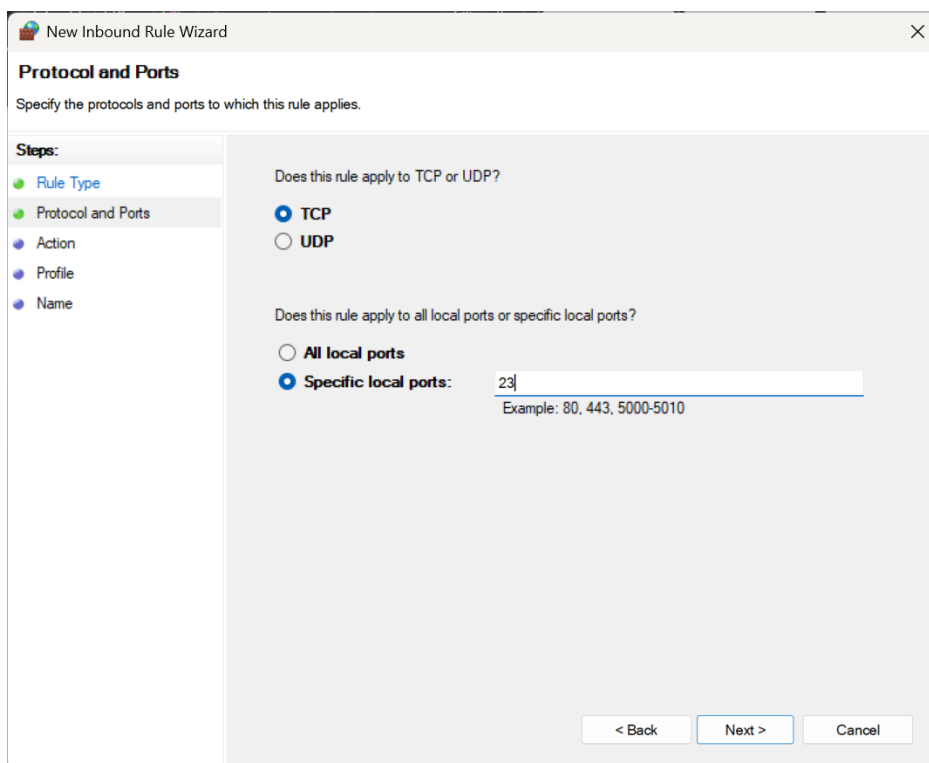
we'll click on "New Rule" in Inbound Rules.

Select "Port" for the rule type.



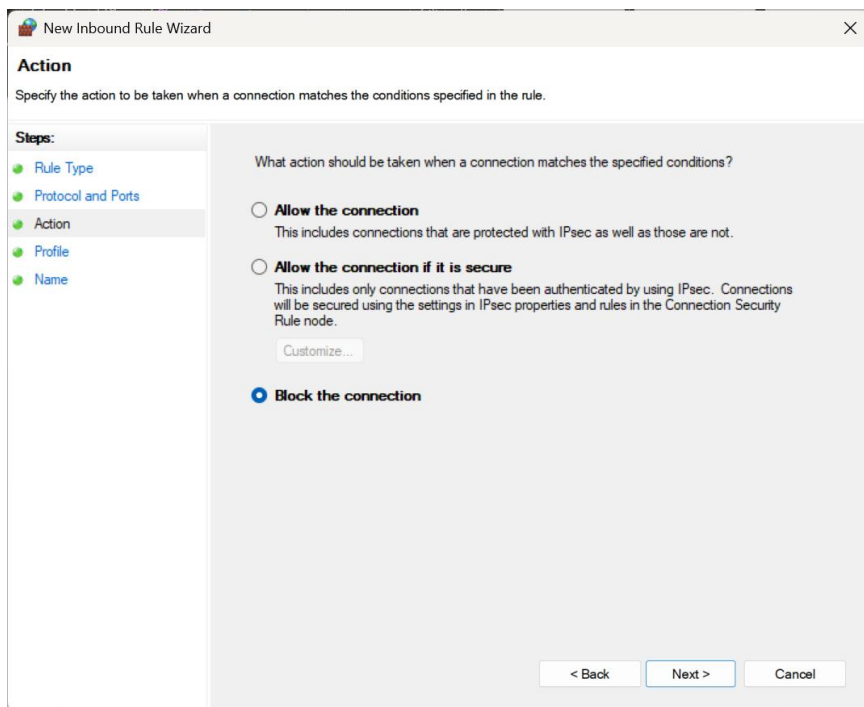
The screenshot shows the 'New Inbound Rule Wizard' window, specifically the 'Rule Type' step. The title bar reads 'New Inbound Rule Wizard'. The main heading is 'Rule Type' with the instruction 'Select the type of firewall rule to create.' On the left, a 'Steps:' sidebar lists 'Rule Type' (selected), 'Protocol and Ports', 'Action', 'Profile', and 'Name'. The main area asks 'What type of rule would you like to create?' and offers four options: 'Program' (Rule that controls connections for a program.), 'Port' (selected, Rule that controls connections for a TCP or UDP port.), 'Predefined:' (with a dropdown menu showing 'AllJoyn Router' and the description 'Rule that controls connections for a Windows experience.'), and 'Custom' (Custom rule.). At the bottom right are buttons for '< Back', 'Next >', and 'Cancel'.

Select "TCP" and enter the port number 23 for Telnet as given in example.



The screenshot shows the 'New Inbound Rule Wizard' window, specifically the 'Protocol and Ports' step. The title bar reads 'New Inbound Rule Wizard'. The main heading is 'Protocol and Ports' with the instruction 'Specify the protocols and ports to which this rule applies.' On the left, a 'Steps:' sidebar lists 'Rule Type', 'Protocol and Ports' (selected), 'Action', 'Profile', and 'Name'. The main area has two questions. The first is 'Does this rule apply to TCP or UDP?' with 'TCP' selected. The second is 'Does this rule apply to all local ports or specific local ports?' with 'Specific local ports:' selected. Below this, a text input field contains '23', with an example 'Example: 80, 443, 5000-5010' shown below it. At the bottom right are buttons for '< Back', 'Next >', and 'Cancel'.

Choose “Block the connection”



New Inbound Rule Wizard

Action

Specify the action to be taken when a connection matches the conditions specified in the rule.

Steps:

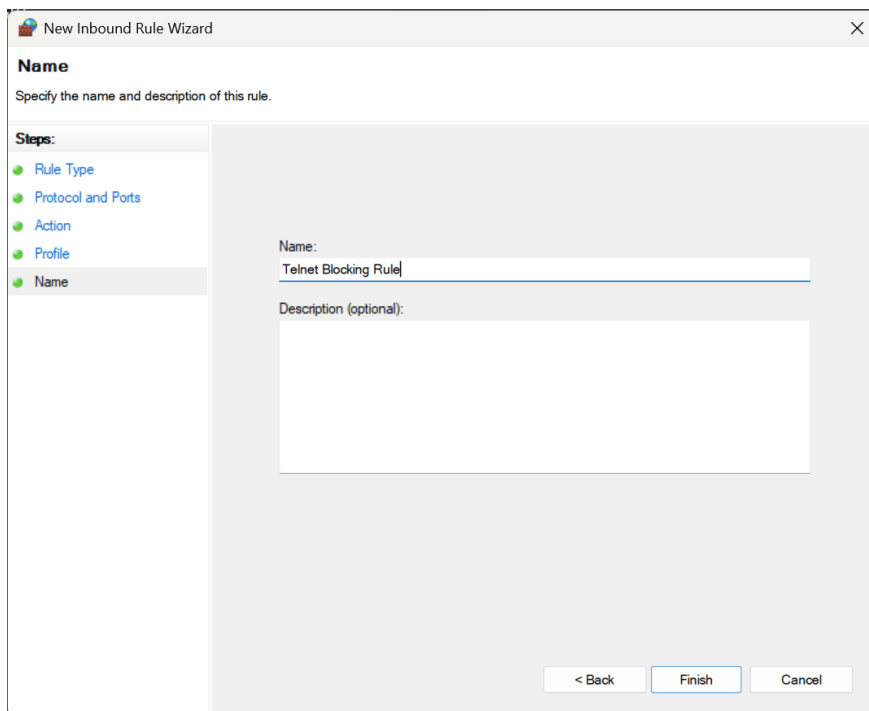
- Rule Type
- Protocol and Ports
- Action**
- Profile
- Name

What action should be taken when a connection matches the specified conditions?

☐ **Allow the connection**
This includes connections that are protected with IPsec as well as those are not.

☐ **Allow the connection if it is secure**
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.

☒ **Block the connection**



New Inbound Rule Wizard

Name

Specify the name and description of this rule.

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name**

Name:

Description (optional):

Let's test the rule

```
Windows PowerShell
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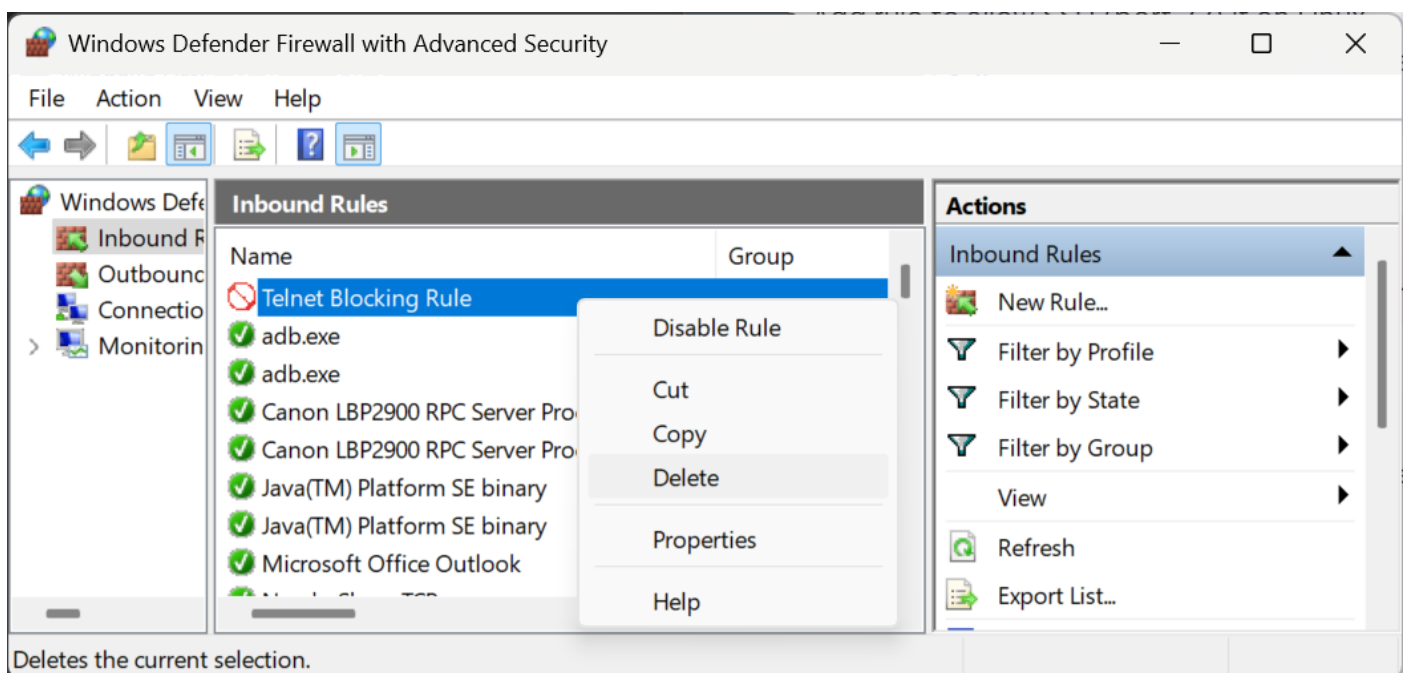
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Admin> Test-NetConnection -ComputerName localhost -Port 23
WARNING: TCP connect to (::1 : 23) failed
WARNING: TCP connect to (127.0.0.1 : 23) failed

ComputerName      : localhost
RemoteAddress     : ::1
RemotePort        : 23
InterfaceAlias    : Loopback Pseudo-Interface 1
SourceAddress     : ::1
PingSucceeded     : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded  : False
```

“TcpTestSucceeded: False” indicates that the rule was successfully added to the firewall.

Now let's delete the same rule.

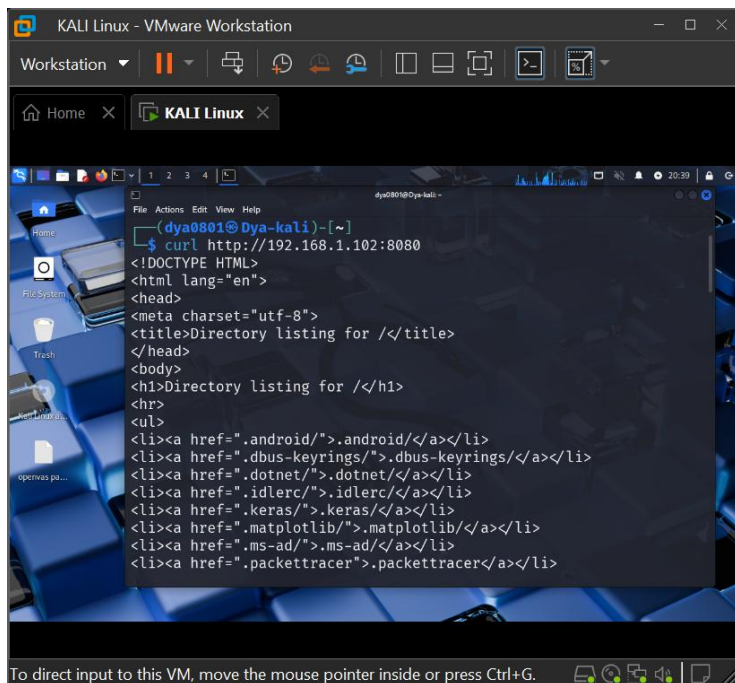


New example, let's test the port 8080(HTTP)

Host a server on 8080 on Windows host.

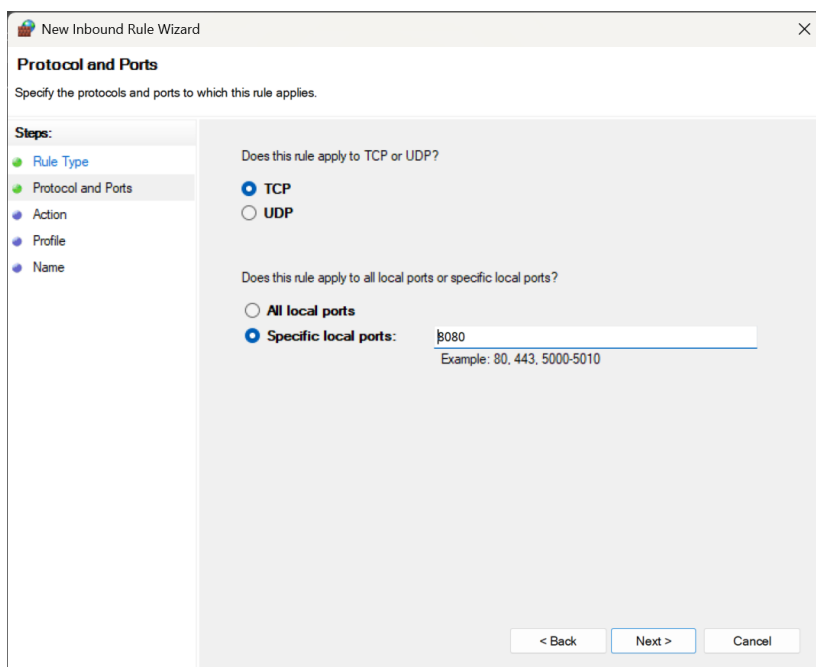
```
PS C:\Users\Admin> cd $env:USERPROFILE
>> python -m http.server 8080
Serving HTTP on :: port 8080 (http://[::]:8080/) ...
```

Sending request to connect to port 8080 from Kali Machine(Virtual Machine).

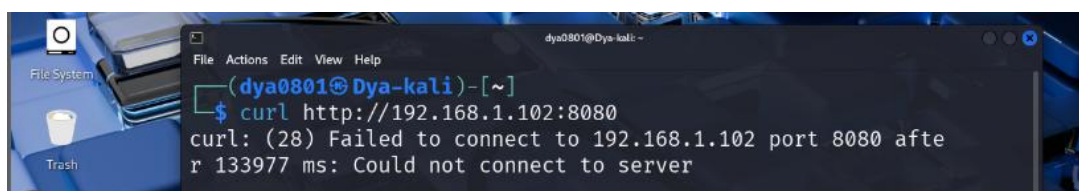


```
(dya0801@Dya-kali)-[~]
$ curl http://192.168.1.102:8080
<!DOCTYPE HTML>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href=".android/">.android/</a></li>
<li><a href=".dbus-keyrings/">.dbus-keyrings/</a></li>
<li><a href=".dotnet/">.dotnet/</a></li>
<li><a href=".idlerc/">.idlerc/</a></li>
<li><a href=".keras/">.keras/</a></li>
<li><a href=".matplotlib/">.matplotlib/</a></li>
<li><a href=".ms-ad/">.ms-ad/</a></li>
<li><a href=".packettracer">.packettracer</a></li>
```

Add an Inbound Rule to block traffic from port 8080(on windows host machine),



Resending request,



```
(dya0801@Dya-kali)-[~]
$ curl http://192.168.1.102:8080
curl: (28) Failed to connect to 192.168.1.102 port 8080 after 133977 ms: Could not connect to server
```